

Nonlinear Systems II: Lyapunov Stability

Proving an equilibrium is stable without ever solving the equations.

Lyapunov's **direct method** certifies stability by finding an energy-like function $V(x) > 0$ that **decreases** along every trajectory ($\dot{V} \leq 0$). If such a V exists, the equilibrium is stable – no closed-form solution required. You will:

- use the pendulum's physical energy as a Lyapunov function and watch it decay along a simulated trajectory, and
- for a linear system, solve the **Lyapunov equation** $A^T P + P A = -Q$ with `lyap` and use $V = x^T P x$.

Run with `publish('LyapunovStability.m')`, or step through with **Ctrl+Enter**.

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Energy as a Lyapunov function (damped pendulum)

For $\ddot{\theta} + b\dot{\theta} + \sin \theta = 0$, the total energy

$$V(\theta, \dot{\theta}) = \frac{1}{2}\dot{\theta}^2 + (1 - \cos \theta) \geq 0$$

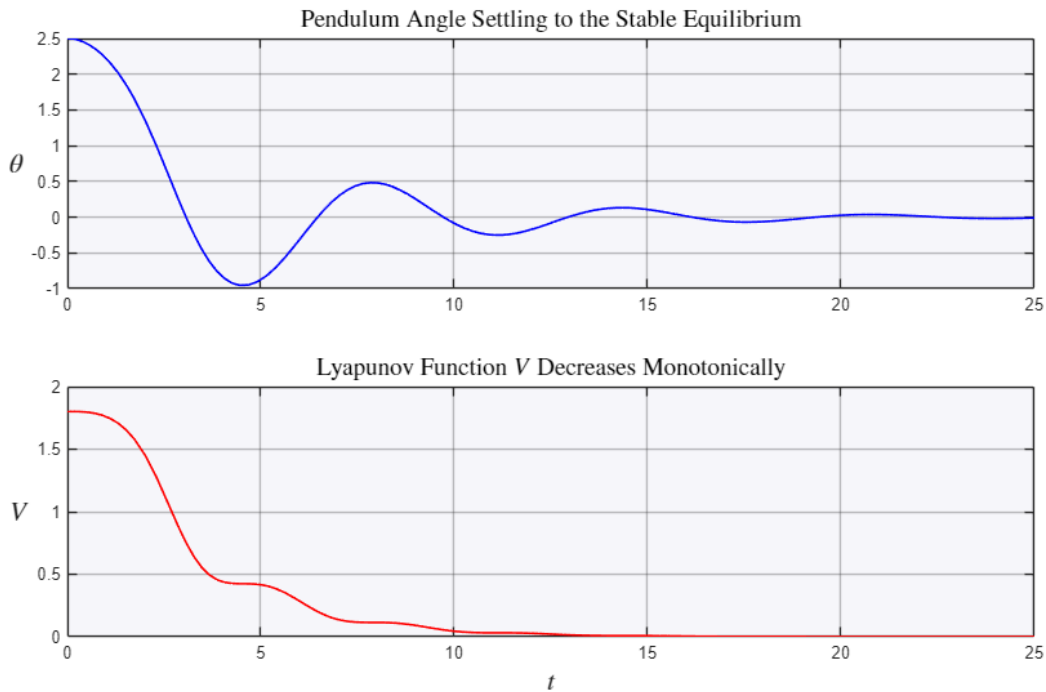
is a natural candidate. Differentiating along trajectories gives $\dot{V} = -b\dot{\theta}^2 \leq 0$: energy can only dissipate, so the hanging-down equilibrium is stable.

```
b = 0.4;
pend = @(t,x)[x(2); -b*x(2) - sin(x(1))];
[t,x] = ode45(pend,[0 25],[2.5; 0]);
V = 0.5*x(:,2).^2 + (1 - cos(x(:,1)));

figure
subplot(2,1,1)
plot(t, x(:,1),'b','LineWidth',1.2)
grid on
title('Pendulum Angle Settling to the Stable Equilibrium','Interpreter','latex','FontSize',14)
ylabel('$\theta$', 'Interpreter','latex','FontSize',16); set(get(gca,'YLabel'),'Rotation',0)
subplot(2,1,2)
plot(t, V,'r','LineWidth',1.2)
grid on
title('Lyapunov Function $V$ Decreases Monotonically','Interpreter','latex','FontSize',14)
ylabel('$V$', 'Interpreter','latex','FontSize',16); set(get(gca,'YLabel'),'Rotation',0)
xlabel('$t$', 'Interpreter','latex','FontSize',16)

fprintf('V(0) = %.4f, V(end) = %.4f (energy dissipated to ~0)\n', V(1), V(end))

V(0) = 1.8011, V(end) = 0.0002 (energy dissipated to ~0)
```



The Lyapunov equation for a linear system

For $\dot{x} = Ax$, a classical theorem says: the origin is asymptotically stable **iff**, for any chosen $Q = Q^T > 0$, the equation

$$A^T P + PA = -Q$$

has a unique solution $P = P^T > 0$. Then $V = x^T P x$ is a Lyapunov function. MATLAB's `lyap(A',Q)` solves exactly this.

```
A = [0 1; -2 -3];
Q = eye(2);
P = lyap(A', Q);
fprintf('\nP =\n'); disp(P)
fprintf('P positive definite? %d (eigenvalues %s)\n', all(eig(P)>0), mat2str(eig(P).',3))
fprintf('Residual ||A\'P + PA + Q|| = %.2e\n', norm(A'*P + P*A + Q))
```

```
P =
    1.2500    0.2500
    0.2500    0.2500
```

```
P positive definite? 1 (eigenvalues [0.191 1.31])
Residual ||A\'P + PA + Q|| = 2.35e-16
```

Visualizing $V = x^T P x$ as a bowl the trajectory rolls down

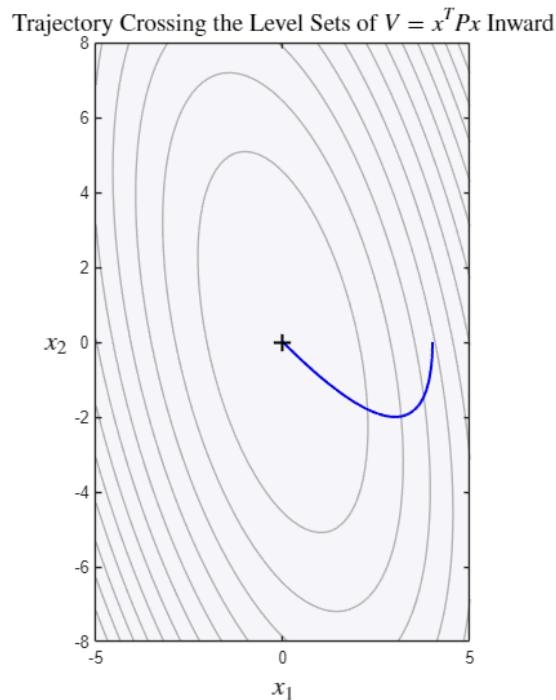
The level sets of $V = x^T P x$ are nested ellipses. A trajectory of $\dot{x} = Ax$ crosses them strictly inward, because $\dot{V} = -x^T Q x < 0$.

```
[t2,x2] = ode45(@(t,x) A*x, [0 8], [4; 0]);
Vlin = sum((x2*P).*x2, 2); % V(t) = x(t)' P x(t)
```

```

[g1,g2] = meshgrid(linspace(-5,5,60), linspace(-8,8,60));
Vgrid = P(1,1)*g1.^2 + 2*P(1,2)*g1.*g2 + P(2,2)*g2.^2;
figure
contour(g1,g2,Vgrid, 12, 'Color',[.7 .7 .7])
hold on
plot(x2(:,1), x2(:,2),'b','LineWidth',1.5)
plot(0,0,'k+','MarkerSize',10,'LineWidth',1.5)
hold off
axis equal
title('Trajectory Crossing the Level Sets of $V=x^TPx$ Inward','Interpreter','latex','FontSize',14)
ylabel('$x_2$','Interpreter','latex','FontSize',16); set(get(gca,'YLabel'),'Rotation',0)
xlabel('$x_1$','Interpreter','latex','FontSize',16)

```



Summary

- Lyapunov's direct method proves stability from an energy function that decreases – no solution of the ODE needed.
- Physical energy is often a ready-made Lyapunov function.
- For linear systems the search is automatic: solve $A^T P + P A = -Q$ with `lyap` and check $P > 0$.

Try it yourself

- Start the pendulum near the top (`[3.0;0]`) and notice V still decreases monotonically – energy dissipates from any initial swing.
- Change Q to `diag([10 1])` and confirm `lyap` returns a different $P > 0$ with the same near-zero residual.